2. Prevention or delay of self – decomposition of the food.
   a. By destruction or inactivation of food enzymes Example: Blanching
   b. By prevention or delay of purely chemical reactions Example: Prevention of oxidation by means of antioxidants.
3. Prevention of damage because of insects, animals, mechanical causes, etc.

5.5 Diary Microbiology

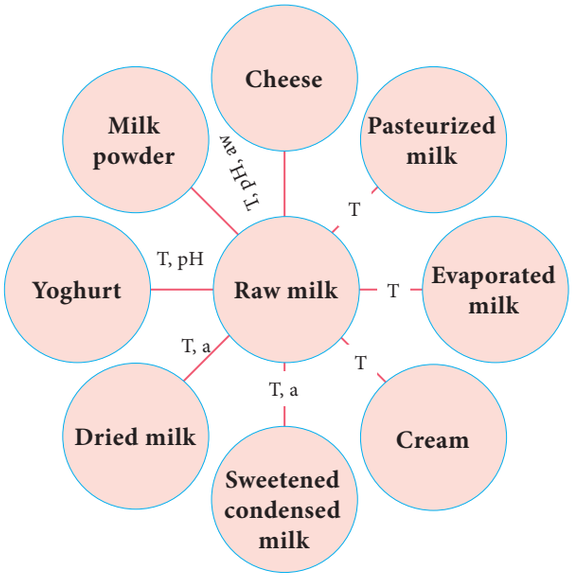
The area of dairy microbiology is large and diverse. The bacteria in dairy products may cause disease or spoilage. Some bacteria may be specifically added to milk for fermentation to produce products like yoghurts and cheese (Figure 5.3).



Figure 5.3: Yoghurts and cheese

5.5.1 MILK

Milk is the fluid, secreted by mammals for the nourishment of their young ones. It is in liquid form without having any colostrum. The milk contains water, fat, protein and lactose. About 80–85% of the protein is casein. Due to moderate pH (6.4–6.6),



Flowchart 5.2: Various products obtained from raw milk

good quantity of nutrients and high water content, milk an excellent nutrient for the microbial growth. The main source of microorganisms under interior eats, surrounding environment and manual milking process, make the source of contamination (Flowchart 5.2).

pH–Hydrogen ion concentration

T–Elevated temperature

H–Reduced water pressure

aw–water activity

5.5.2 Composition and Properties

Milk is considered to be the "Most nearly perfect" food for man and hence is one of the most important ingredients of the diet. It is an extremely complex mixture and usually contains (Table 5.3).

5.5.3 Sources of Microorganisms in Milk

- Three sources contribute to the microorganism found in milk the udder interior, the teat exterior and

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